Guangdong Technion
Israel Institute of Technology
广东以色列理工学院

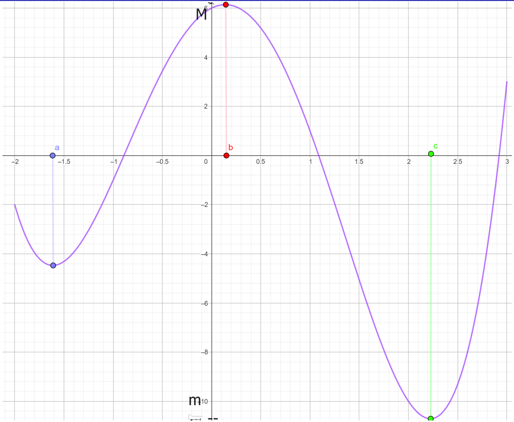
GTIIT

Calculus 1 - M
Winter Semester - 2024

Lecture 11 - November 18th

Applied Project: Rainbows. Perhaps you have seen a secondary rainbow above the primary bow. That results from the part of a ray that enters a raindrop and is refracted again. Show that the colors in the secondary rainbow appear in the opposite order from those in the primary rainbow. (Stewart, p 285).

Absolute Extreme Values



Ex. 1 .

- $M = f(b)$ is the absolute maximum value of f on $(-2, 3)$
- $m = f(c)$ is the absolute mimimum value of f on $(-2, 3)$

Absolute Extreme Values

Let D be the domain of f and $c \in D$, then

• $f(c)$ is the absolute maximum value of f on D if $f(c) \geq f(x)$ for all $x \in D$.

• $f(c)$ is the absolute mimimum value of f on D if $f(c) \leq f(x)$ for all $x \in D$.

An absolute maximum or minimum is sometimes called a global maximum or minimum.

Absolute Extreme Values

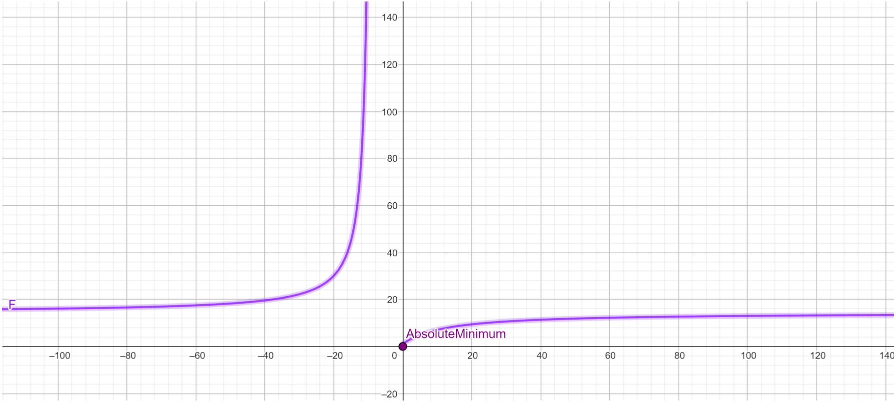
- The maximum and minimum values of f are called the extreme values of f on D.

- If $f(b)$ is the absolute maximum value then $(b, f(b))$ is highest point on the graph;

- If $f(c)$ is the absolute minimum value, then $(c, f(c))$ is lowest point.

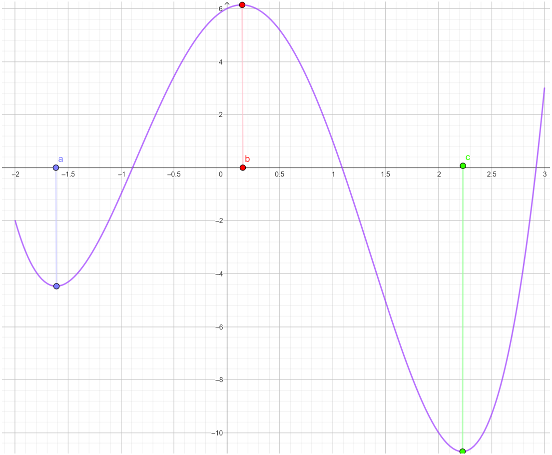
Absolute Extreme Values do not always exist

Ex. 2



- $0 = f(0)$ is the absolute minimum value of f on $Dom(f)$
- f has no absolute maximum on $Dom(f)$

Ex. 3



- $f(a)$ is a local minimum value of f on $(-2,3)$
- $f(b)$ is a local (and absolute) maximum value of f on $(-2,3)$

Local Extreme Values

Let D be the domain of f and $c \in D$. Then

- $f(c)$ is a local maximum value of f if $f(c) \geq f(x)$ for all x near c.

- $f(c)$ is a local mimimum value of f if $f(c) \leq f(x)$ for all x near c.

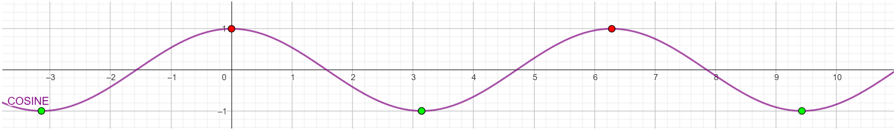
An absolute maximum or minimum is also called a relative maximum or minimum.

Local Extreme Values

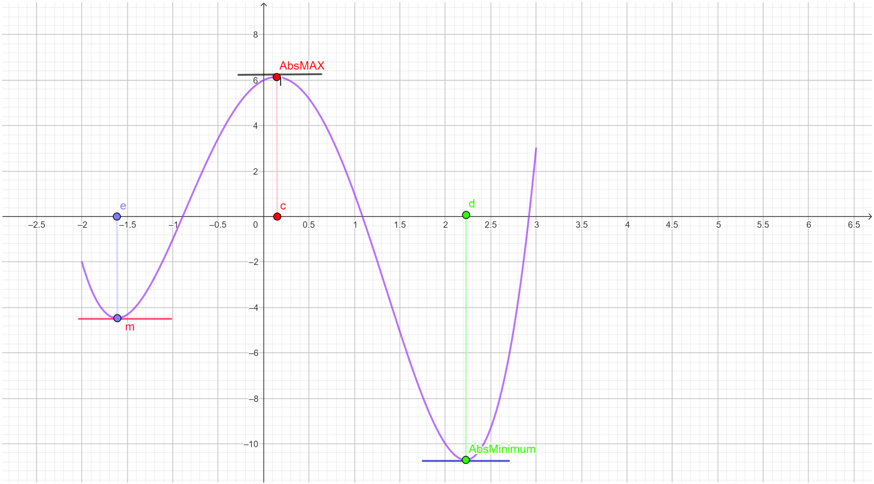
Ex. 4

- $\cos(x)$ takes on its local and absolute maximum value 1 at $2n\pi\ \forall n \in \mathbb{Z}$ since $f(2n\pi) = 1$

- $\cos(x)$ takes on its local and absolute minimum value -1 at $(2n+1)\pi\ \forall n \in \mathbb{Z}$ since $f((2n+1)\pi) = -1$



Fermat's Theorem



Fermat's Theorem: main tool to find the local extreme values

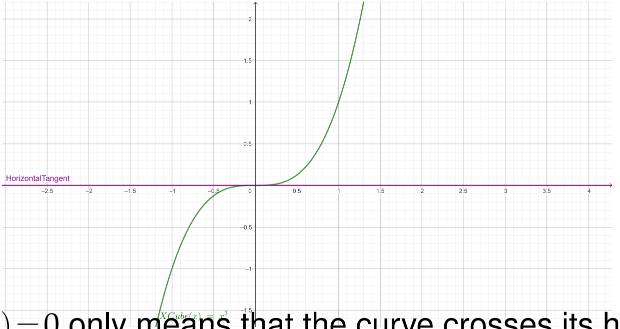
If f has a local maximum or minimum at c, and $f'(c)$ exists, then $f'(c) = 0$.

Be careful: the converse of Fermat's Theorem is false in general.

Ex. 5. a) Be careful when using Fermat's Theo: consider $f(x) = x^3$

- $f'(x) = 3x^2$, so $f'(0) = 0$.
- f has no maximum or minimum at $x = 0$, because:

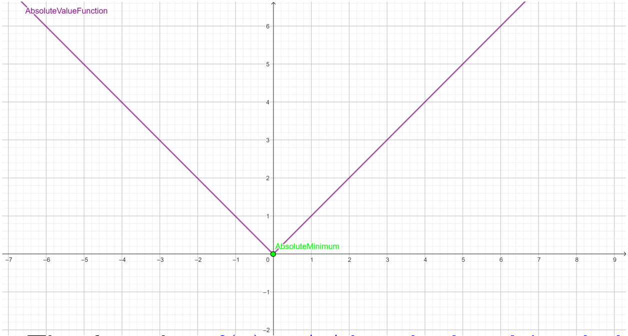
$$x^3 < 0 \;\forall\; x < 0 \text{ and } 0 < x^3 \;\forall\; 0 < x$$



- $f'(0) = 0$ only means that the curve crosses its horizontal tangent at $(0, 0)$.

Ex. 5. b) Be careful when using Fermat's Theorem

- $f'(0)$ does not exists.



- The function $f(x) = |x|$ has its local (and absolute) minimum value $0 = f(0)$, because

$$0 < |x| \ \forall \ x \neq 0$$

Ex. 5. c) Prove that $f(x) = x^{2/3}$ has a local and absolute minimum value at $x = 0$.
Is there any contradiction with Fermat's Theorem?

- The function $f(x) = x^{2/3}$ has its local (and absolute) minimum value $0 = f(0)$, because, for any $x \neq 0$, the value of $f(x)$ is strictly positive, since

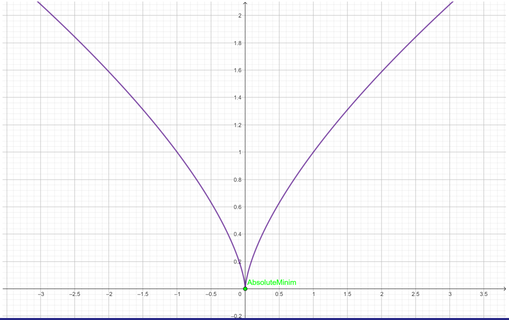
$$\forall\, x \neq 0,\ f(x) = x^{2/3} = \left(x^{1/3}\right)^2 > 0$$

Hence, by definition, $f(x) = x^{2/3}$ has its local minimum value at $0 = f(0)$ and also absolute minimum value .

$\bullet\ f'(x) = \frac{2}{3}x^{-1/3}\ \forall\ x \neq 0$

- There is no contradiction with Fermat's Theorem, since $f'(0)$ does not exists:

$$f'(0) = \lim_{x \to 0} \frac{f(x) - f(0)}{x} = \lim_{x \to 0} \frac{x^{2/3}}{x} = \lim_{x \to 0} \frac{2}{3x^{1/3}} = \infty \checkmark$$



Critical Point

A point c in the domain of f is a critical point if either $f'(c) = 0$ or $f'(c)$ does not exist.

Fermat's Theorem says then:

- If f has a local maximum or minimum at c, then

$\boxed{c \text{ is a critical point of } f.}$

- Fermat's Theorem does suggest that we should at least start looking for extreme values of f at the points c where $f'(c) = 0$ or $f'(c)$ does not exist:

these candidates to be $\boxed{\text{local maximum or minimum}}$ are called $\boxed{\text{critical points}}$

Fermat's Theorem: main tool to find the local extreme values

Ex. 6. a) Find the local extreme values of $f(x) = x^{3/5}(4-x)$.

Solution. In virtue of Fermat's Theorem, let us find the critical points, for which it is necessary to compute its derivative:

$$\boxed{f'(x)} = \frac{3}{5}x^{-2/5}(4-x) - x^{3/5} = \boxed{\frac{4(3-2x)}{5x^{2/5}} \quad \forall\, x \neq 0}$$

But $\boxed{f'(0) \text{ does not exists:}}$

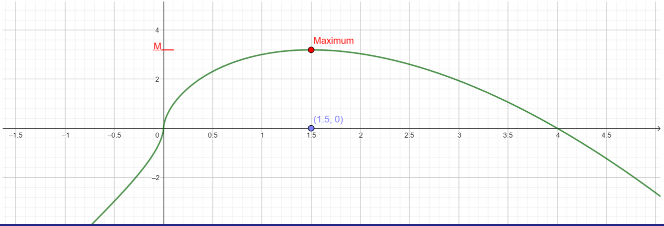
$$\boxed{f'(0)} = \lim_{x\to 0}\frac{f(x)-f(0)}{x} = \lim_{x\to 0}\frac{\frac{3}{5}x^{-2/5}(4-x) - x^{3/5}}{x} = \infty$$

Therefore, f has two critical points:

- $x = 0$, where the derivative does not exists, and $x = 0$ is not a local maximum or minimum

- $x = \frac{3}{2}$, where $f'(\frac{3}{2}) = 0$

- f has a local (and also absolute) maximum value at $x = \frac{3}{2}$, where $f'(\frac{3}{2}) = 0$, whith value $M = f(\frac{3}{2}) = \frac{5}{2}(\frac{3}{2})^{3/5}$ ✓



Increasing and Decreasing Test

1. If $f'(x) > 0$ for all $x \in (a, b)$, then f is increasing on (a, b).
2. If $f'(x) < 0$ for all $x \in (a, b)$, then f is decreasing on (a, b).

First Derivative Test.

Suppose that c is a critical point of a continuous f.

1. If f' changes from positive to negative at c, then f has a local maximum at c.
2. If f' changes from negative to positive at c, then f has a local minimum at c.
3. If f' is positive to the left and right of c, then f has no local maximum or minimum at c.
4. If f' is negative to the left and right of c, then f has no local maximum or minimum at c.

Ex. 6. b) Use the Increasing and Decreasing Test to justify that $f(x) = x^{3/5}(4 - x)$ has a local maximum at $x = \frac{3}{2}$.

Solution. $\boxed{f'(x) = \dfrac{4(3 - 2x)}{5x^{2/5}} \quad \forall\, x \neq 0}$. Let us study the intervals where $f' > 0$ and those where $f' < 0$:

- Denominator of f' is always positive, because it is a square: $5x^{2/5} > 0 \ \ \forall\, x \neq 0$.
- Numerator: $3 - 2x > 0 \iff \frac{3}{2} > x \iff x \in (-\infty, \frac{3}{2})$
- Numerator: $3 - 2x < 0 \iff \frac{3}{2} < x \iff x \in (\frac{3}{2}, \infty)$

	$(-\infty, \frac{3}{2})$	$\frac{3}{2}$	$(\frac{3}{2}, +\infty)$
Numerator	$+$	0	$-$
Denominator	$+$	$\neq 0$	$+$
$f'(x)$	$+$	0	$-$

We conclude then

1. $f'(x) > 0$ on $(\frac{3}{2}, \infty)$, then f is increasing on $(\frac{3}{2}, \infty)$.
2. $f'(x) < 0$ on $(-\infty, \frac{3}{2})$, then f is decreasing on $(-\infty, \frac{3}{2})$

$\Rightarrow f$ has a local maximum value at $x = \frac{3}{2}$ ✓
In fact, f reaches the absolute maximum value at $x = \frac{3}{2}$
with $M = f(\frac{3}{2}) = (\frac{3}{2})^{3/5}(\frac{5}{2})$ ✓

Absolute Extreme Values exist for

f continuous

on a closed interval $[a, b]$

The Extreme Value Theorem

If f is continuous on a closed interval $[a, b]$, then

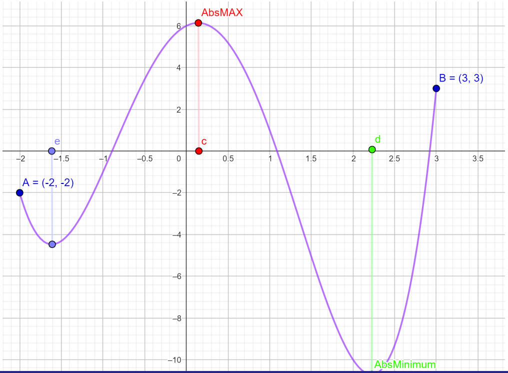
- f attains an absolute maximum value $M = f(c)$, and

- f attains an absolute minimum value $m = f(d)$, at some numbers $c, d \in [a, b]$.

Absolute Extreme Values exist for f continuous on a closed interval $[a, b]$

Ex. 7

- $f(c)$ is the absolute maximum on $[-2, 3]$
- $f(d)$ is the absolute minimum value of f on $[-2, 3]$



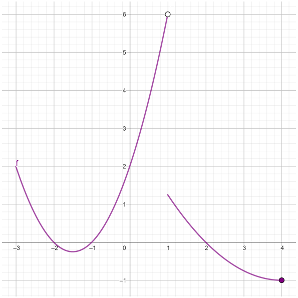
Hypothesis of the Extreme Value Theorem: continuity on a closed interval

- If the hypothesis are not verified, the theorem does not hold (it is not necessarily true)

- If f is not continuous or the domain is an open interval (a, b), then the hypothesis of the theorem are not verified and f may not attain absolute extreme values.

- Nonetheless, a discontinuous function could have maximum and minimum values.

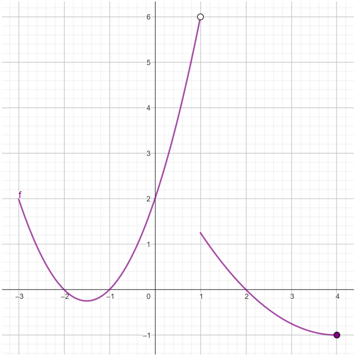
Ex. 8 $y = f(x)$ is discontinuous



- The range of f is $[-1, 6)$.

- $f(x) < 6\ \forall x$ and the function takes on values arbitrarily close to 6, but never actually attains the value 6.
- This does not contradict the Extreme Value Theorem because f is not continuous.

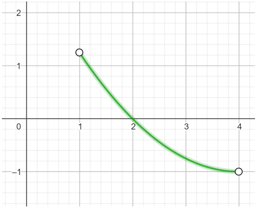
$y = f(x)$ is discontinuous



- f has the absolute minimum value $-1 = f(4)$

- f has no absolute maximum value, because $(1, 6) \notin$ Graph(f)

$y = g(x)$: g is defined on the open interval $(1, 4)$



- Range$(f) = (-1,\ 1.2)$ but $y = -1$ and $y = 1.2$ are not attained

- g has no absolute minimum value, since $(4,\ -1) \notin$ Graph(f),
  because $x = 4 \notin$ Dom(f)

- g has no absolute maximum value, since $(1,\ 1.2) \notin$ Graph(f),
  because $x = 1 \notin$ Dom(f)

Next Saturday, November 23rd
Midterm Exam

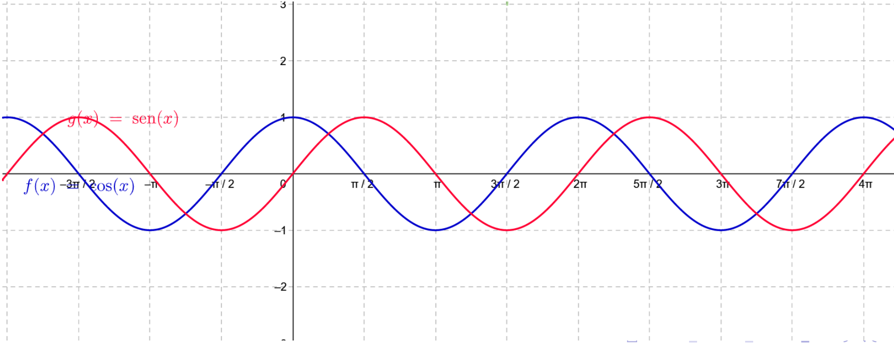
- Topics: include up to Chapter 3 (Stewart) and all our lectures, tutorials and workshops up to November 15th.
- Concepts and Theorems given this week are NOT INCLUDED

Short Review: Remember

The following limits do not exist:

- $\lim_{x\to\infty} \sin x$ does not exist

- $\lim_{x\to\infty} \cos x$ does not exist.



Short Review: Compute the following limits

a) $lim_{x\to 0}\left[x^2 \cdot \left(\sin(\frac{1}{x}) + 8\cos(3x)\right)\right]$

b) Does f has a removable discontinuity at $x = 0$? Justify.

c) $lim_{x\to+\infty}\left[x \cdot \sin(\frac{1}{x})\right]$

d) $lim_{x\to+\infty}\left[\frac{4}{x^3} \cdot \sin(\frac{1}{x})\right]$

Solutions. a) $L = 0$, c) $L = 1$, d) $L = 0$.

Task: Prove (a) and (d) by definition!

Solution. Use the Squeeze Theorem to compute the limit

$$a)\ 0 \le \left|x^2\cdot\left(\sin(\tfrac{1}{x})+8\cos(3x)\right)\right| = |x|^2\cdot\underbrace{\left|\sin\left(\tfrac{1}{x}\right)+8\cos(3x)\right|}_{\le 1+8} \le 9|x|^2$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ 0 & L & 0 \end{array}$$

$$\Rightarrow \lim_{x\to 0}\left[x^2\cdot\left(\sin(\tfrac{1}{x})+8\cos(3x)\right)\right] = L = 0\ \checkmark$$

Then, the limit exists and so,

f has removable discontinuity at $x=0$ ✓

Solution.

Use the Composition Rule and the limit

$$\lim_{t\to 0}\frac{\sin t}{t} = 1$$

c) Limit at ∞: $lim_{x\to+\infty}\left[x\cdot\sin(\frac{1}{x})\right]$.
Take $t = \frac{1}{x} \to 0$ as $x \to +\infty$, then

$$\lim_{x\to+\infty}\left[x\cdot\sin(\tfrac{1}{x})\right] = \lim_{x\to+\infty}\left[\frac{\sin(\frac{1}{x})}{\frac{1}{x}}\right] = \boxed{\lim_{t\to 0}\frac{\sin t}{t} = 1 \checkmark}$$

Solution. Use the Squeeze Theorem

d) Limit at ∞:

$$0 \leq \left| \frac{4}{x^3} \cdot \sin\left(\frac{1}{x}\right) \right| = \left|\frac{4}{x^3}\right| \cdot \underbrace{\left|\sin\left(\frac{1}{x}\right)\right|}_{\leq 1} \leq \left|\frac{4}{x^3}\right|$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ 0 & L & 0 \end{array}$$

$$\Rightarrow \boxed{\lim_{x\to+\infty}\left[\frac{4}{x^3}\cdot\sin\left(\frac{1}{x}\right)\right] = L = 0 \checkmark}$$

The Closed Interval Method to find the absolute maximum and minimum values

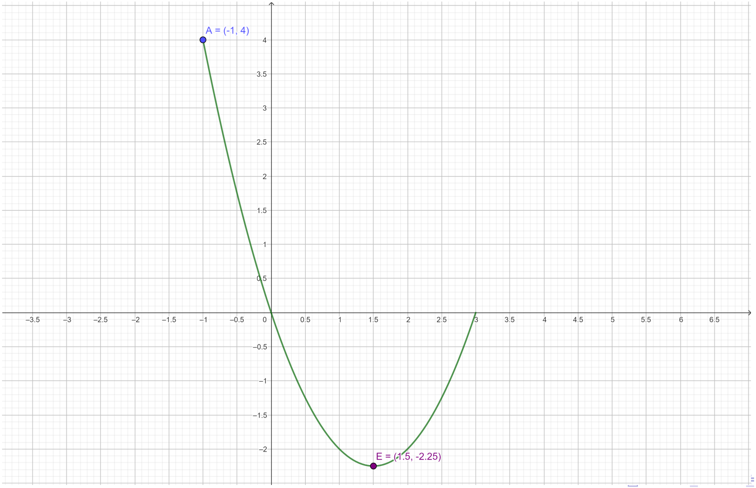
Let f be a continuous function defined on a closed interval $[a, b]$. Proceed as follows to determine the absolute extreme values:

1. Find the critical points of f in the open (a, b)
2. Find the values of f at each of the critical points
3. Find the values of f at the endpoints of the interval a, b.
4. • The largest of the values from Steps 2 and 3 is the absolute maximum value;
   • the smallest of these values is the absolute minimum value.

Absolute Extreme Values exist for f continuous on a closed interval $[a, b]$

Ex. 9. Graph f and find the absolute extreme values of $f(x) = x^2 - 3x$ on $[-1, 3]$.
Are they attained on interior points or at any of the endpoints of the closed interval?

Absolute Extreme Values exist for f continuous on a closed interval $[a, b]$



Find the critical points of f on $[-1,3]$ and compare all the values with f on -1 and 3:

•f is differentiable on $\mathbb{R}$, so the only critical points are those x where $f'(x) = 0$.

• $f(x) = x^2 - 3x \Rightarrow f'(x) = 2x - 3$, then

$$\boxed{f'(x) = 0 \Rightarrow x = 3/2}$$

Compare the three values of $f(x)$ on these 3 points:

• $f(3/2) = (x^2 - 3x)\Big|_{x=3/2} = -9/4$

• $f(-1) = (x^2 - 3x)\Big|_{x=-1} = 4$

• $f(3) = (x^2 - 3x)\Big|_{x=3} = 0$

f attains the absolute maximum and the absolute minimum on $[-1, 3]$

- $4 = f(-1)$ is the absolute maximum on $[-1, 3]$ and it is attained at the left endpoint of the closed interval

- $f(3/2) = -9/4$ is the absolute minimum value of f on $[-1, 3]$ and it is attained at an interior point

$$\boxed{f'(3/2) = 0\checkmark}$$

Ex. 10. Follow the Closed Interval Method to find the absolute maximum and minimum values of $f(x) = \dfrac{x-1}{(x^2-x+1)}$ on $[-1, 4]$

$f(x)$ is continuous and defined on a closed interval:

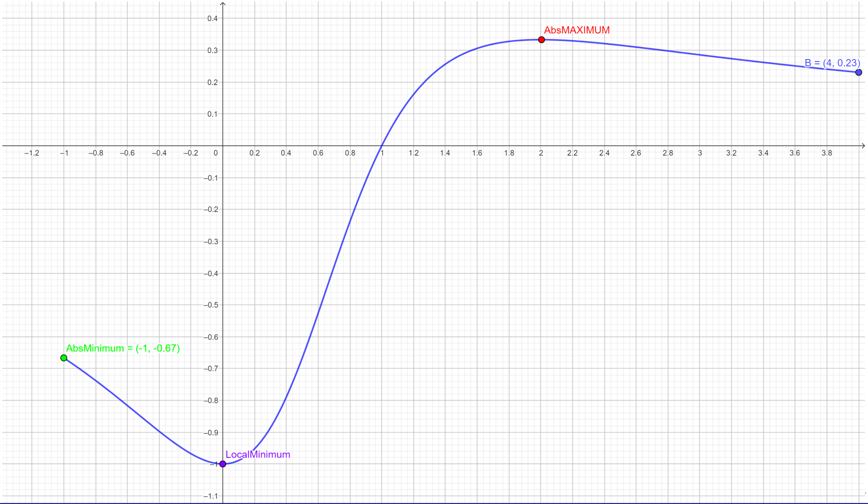
$$f(x) = \frac{x-1}{(x^2-x+1)} = \frac{x-1}{((x-\frac{1}{2})^2+\frac{3}{4})}$$

1. $f(x)$ is differentiable on $(-1, 4)$, so critical points are

$$f'(x) = -\frac{(x-2)x}{(x^2-x+1)^2} = 0 \Rightarrow x = 0 \;\; and \;\; x = 2$$

2. Compute the values of f at each of the critical points
3. Compute the values of f at the endpoints of $[-1, 4]$.

Absolute Extreme Values exist for f continuous on a closed interval $[a, b]$



Compare the values of $f(x)$ at these 4 numbers: the critical points and the two endpoints of the interval

- $x = 0 \in (-1, 4)$ is a critical point
- $x = 2 \in (-1, 4)$ is a critical point
- $x = -1$ is the left endpoint of $[-1, 4]$
- $x = 4$ is the right endpoint of $[-1, 4]$

Compare the 4 numbers: the highest value is the absolute maximum and the lowest is the absolute minimum of f on $[-1, 4]$

- $f(0) = -1$
- $f(2) = \frac{1}{3}$
- $f(-1) = -\frac{2}{3}$
- $f(4) = \frac{3}{13}$

- $\boxed{-\frac{2}{3} = f(-1)}$ is the absolute minimum on $[-1, 4]$ and it is attained at the left endpoint of the closed interval
- $\boxed{1/3 = f(2)}$ is the absolute maximum value of f on $[-1, 4]$ and it is attained at an interior point ✓

Rolle's Theorem

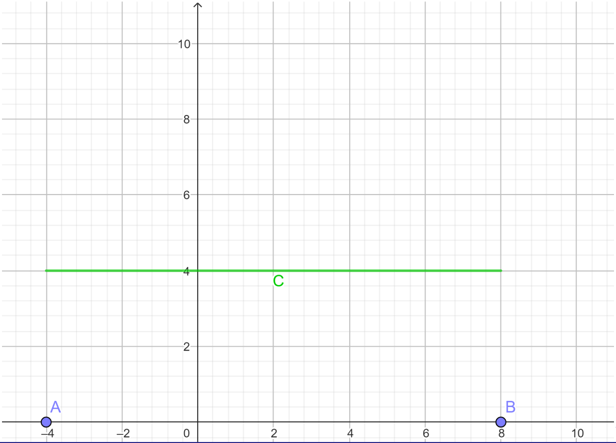
Let f be a function that satisfies the following three hypotheses:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b).
3. $f(a) = f(b)$.

Then there is a number $c \in (a, b)$ such that $f'(c) = 0$.

Let's take a look at the graphs of some typical functions that satisfy the three hypotheses.

$y = C$



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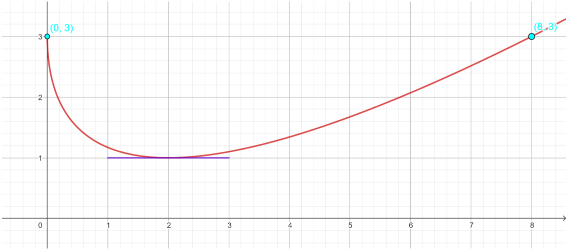
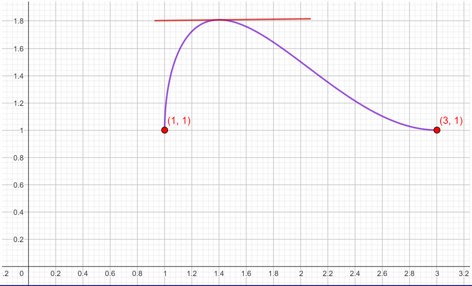
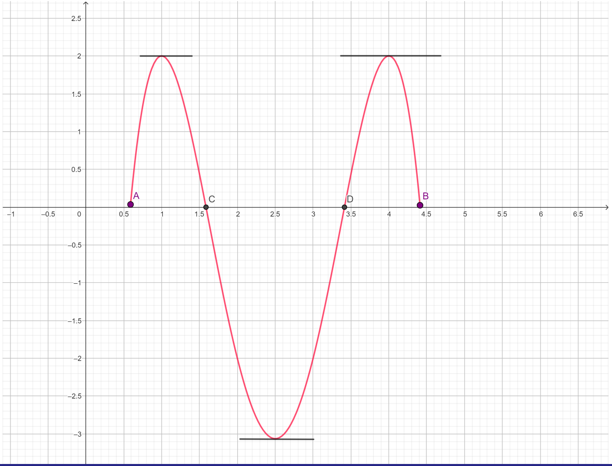




If the three hypotheses are satisfied in more intervals, then there are more points with horizontal tangent line



Ex. 11. Prove that the equation $x^3 + x - 1 = 0$

has exactly one real root.

Solution.

1. First we use the Intermediate Value Theorem to show that at least one root exists:

$$f(0) = -1 < 0 \ \ and \ \ 0 < 1 < f(1)$$

   Since f is continuous on $\mathbb{R}$, the Intermediate Value Theorem states that there is $a \in (0,1) : f(a) = 0$

2. To prove that f has only one root, we use Rolle's Theorem:
   On the contrary, suppose that there are two roots

- Suppose, then, that there are two numbers a, b such that $f(a) = 0 = f(b)$.
- The hypotheses of Rolle's Theorem are verified on $[a, b] \Rightarrow$ there exists c between a and b such that $f'(c) = 0$.
- But this leads to a contradiction, because:

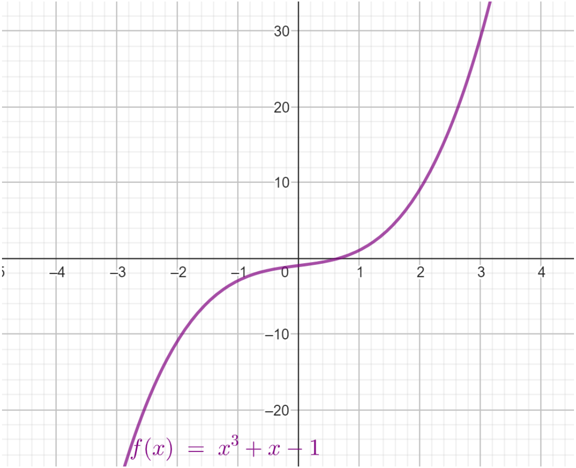
$$\boxed{f'(x) = 3x^2 + 1 \geq 1 > 0}$$

Then, the number c between a and b sucht that $f'(c) = 0$, can not exists.

Hence,

$\boxed{\text{the equation has only one root } a \in (0,1) : f(a) = 0 \checkmark}$

The equation has only one root $a \in (0,1) : f(a) = 0$



The Mean Value Theorem.

Let f be a function that satisfies the following hypotheses:

1. f is continuous on the closed interval $[a, b]$
2. f is differentiable on the open interval (a, b).

Then there is a number c in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

The Mean Value Theorem says that there is at least one point $P = (c, f'(c))$ on the graph where the slope of the tangent line is the same as the slope of the secant line AB

The Mean Value Theorem.

The tangent line at P is parallel to the secant line joining $A = (a, f(a))$ and $B = (b, f(b))$.

Theorem: Slope $= 0$ on (a, b) means f is constant on (a, b)

If $f'(x) = 0$ for all x in an open interval (a, b), then f is constant on (a, b).

Proof:

- Let $x_1, x_2 \in (a, b)$ be any two different numbers: $x_1 < x_2$.
- Since f is differentiable on (a, b), it must be differentiable on (x_1, x_2) and continuous on $[x_1, x_2]$.
- By applying the Mean Value Theorem to f on the interval (x_1, x_2), we get a number $c : \; x_1 < c < x_2$ sucht that

$$\frac{f(x_2) - f(x_1)}{(x_2 - x_1)} = f'(c) = 0 \Rightarrow f(x_2) - f(x_1) = \underbrace{f'(c)}_{=0}(x_2 - x_1)$$

$$\Rightarrow f(x_2) - f(x_1) = 0$$

Hence, $\boxed{f(x_1) = f(x_2).}$

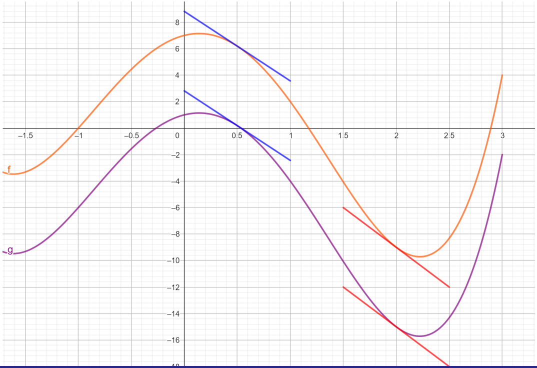
But this is true for any pair of numbers, then f is constant on (a, b) ✓

Corollary: Same slope on (a, b) means $f - g$ is constant

If $f'(x) = g'(x)$ for all x in an open interval (a, b), then $f - g$ is constant on (a, b).

That is, there is a c such that, for all x in (a, b),

$$\boxed{f(x) = g(x) + c \text{ for all } x \in (a, b) \checkmark}$$



Proof:

Apply the previous theorem to $h = f - g$, since:

$$h' = f' - g' = 0 \Rightarrow f - g = h = c \in \mathbb{R}$$

Hence, there exists a constant $c \in \mathbb{R}$ such that for any $x \in \mathbb{R}$:

$$f(x) - g(x) = c \checkmark$$

Ex. 12. Suppose that $11 \leq 4 \cdot f'(x) - 1 \leq 19$ for all x. Show that then $18 \leq f(8) - f(2) \leq 30$

Solution. By hypothesis:

- $11 \leq 4 \cdot f'(x) - 1 \leq 19 \Rightarrow \boxed{3 \leq f'(x) \leq 5 \ \forall\ x}$

- f is differentiable $\forall\ x$, so by the Mean Value Theorem on $[2, 8]$, there is c such that $2 < c < 8$ and

$$f'(c) = \frac{f(b) - f(a)}{b - a} \Rightarrow 3 \leq \frac{f(b) - f(a)}{b - a} < 5$$

This is true for $[a, b] = [2, 8]$

$$\Rightarrow 3 \leq \frac{f(8) - f(2)}{8 - 2} < 5 \Rightarrow \boxed{18 \leq f(8) - f(2) < 30 \ \checkmark}$$

Ex. 13. Suppose that f and g are continuous on $[a,b]$ and differentiable on (a,b). Suppose also that $g(a) = f(a)$ and $g'(x) < f'(x)$ for all $a < x < b$.

Prove that $g(b) < f(b)$.

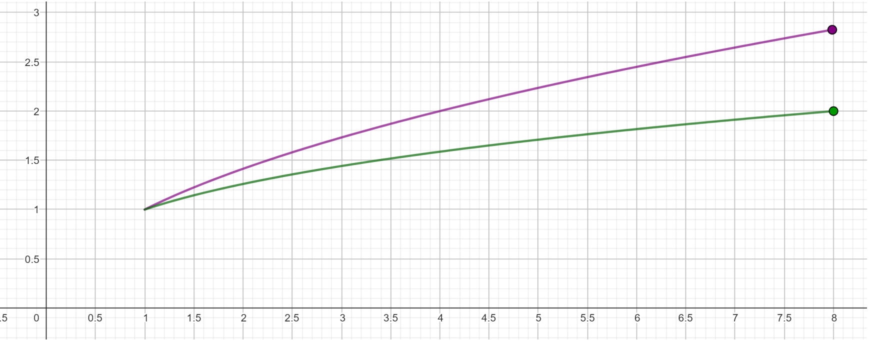
Proof. Apply the previous theorem to $h = f - g$, since:

$$h' = f' - g' > 0 \Rightarrow f - g \text{ is increasing}$$

Hence, $a < b \Rightarrow 0 = (f-g)(a) < (f-g)(b)$

$$g(b) < f(b) \checkmark$$

$0 = h(a)$ and h is increasing $\Rightarrow 0 < h(b)$ ✓

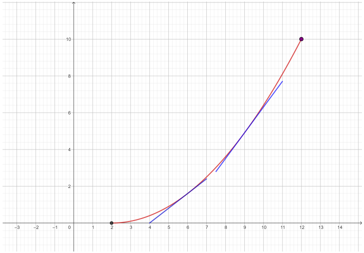


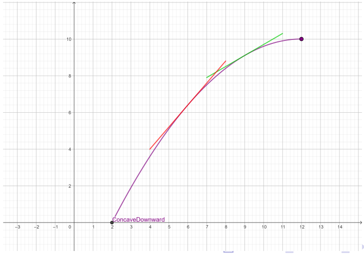
$$h(x) = f(x) - g(x)$$

What Does f'' Say About f?

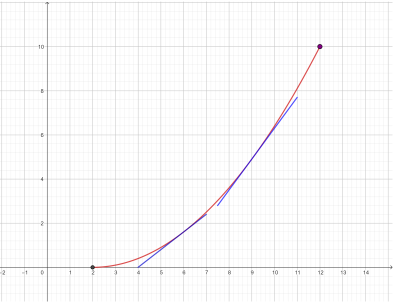
The figures show the graphs of two increasing functions on $[a, b]$, but they look different because they bend in different directions: the concavity is different between them.

$\Rightarrow f''$ allows to distinguish between these two types of behavior





1. If the graph of f lies above all of its tangents on an interval (a, b), then it is called concave upward on (a, b).
2. If the graph of f lies below all of its tangents on (a, b), it is called concave downward on (a, b).



Concave Upward

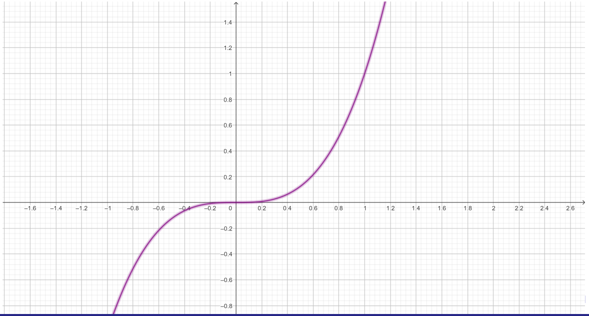
Concave Downward

Concavity and the Second Derivative

Suppose f'', the 2nd derivative of f, exists on (a,b), then

1. If $f'' > 0$ on (a,b) then f has positive concavity on (a,b).
2. If $f'' < 0$ on (a,b) then f has negative concavity on (a,b).

Negative Concavity on $(-\infty, 0)$ Positive Concavity on $(0, +\infty)$



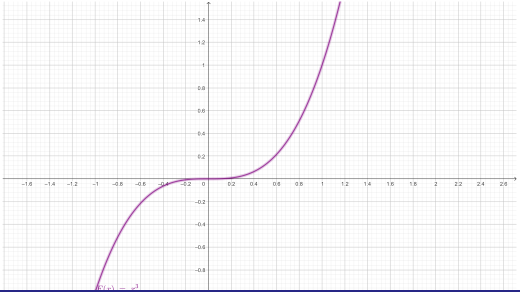
A point $(c, f(c))$ on a curve $y = f(x)$ is called an inflection point if f is continuous at c and:

- the curve changes from concave upward to concave downward at c, or
- the curve changes from concave downward to concave upward at c.

Ex: $f(x) = x^3$ has an inflection point at $x = 0$:

Negative Concavity on $(-\infty, 0)$

Positive Concavity on $(0, +\infty)$



The Second Derivative Test

Suppose f'', the 2nd derivative of f, is continuous on (a, b) and $c \in (a, b)$.

1. If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c.
2. If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c.

- The Second Derivative Test is inconclusive when $f''(0) = 0$: at such a point there might be a maximum, there might be a minimum, or there might be neither.

- This test also fails when $f''(0) = 0$ does not exist.

Example. For $f(x) = x^4 - 4x^3$, analize f' and f'' and find:

1. Intervals of increase and decrease of f.
2. Local maximum and minimum points.
3. Intervals of positive and negative concavity of f.
4. Inflection points.
5. All the asymptotes.
6. Use this information to sketch the curve.

Solution. • f is differentiable on $\mathbb{R}$ because it is a polymonial, hence, 1) and 2) will be analized through f':

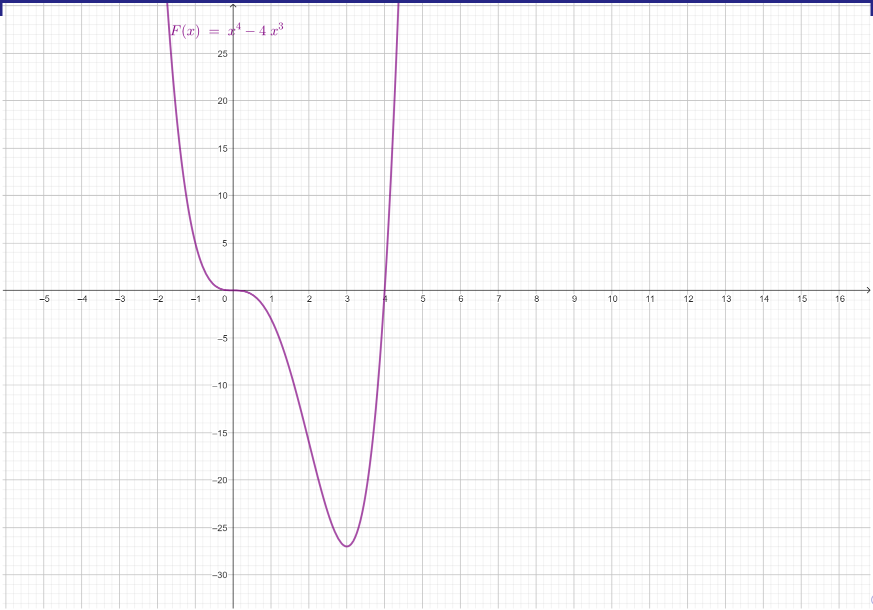
$$\boxed{f'(x)} = 4x^3 - 12x^2 \ \boxed{= 4x^2(x-3)}$$

Then, the critical points are $x = 0$ and $x = 3$

	$(-\infty, 0)$	0	$(0, 3)$	3	$(3, \infty)$
x^2	$+$	0	$+$	$+$	$+$
$(x-3)$	$-$	$-$	$-$	0	$+$
$f'(x)$	$-$	0	$-$	0	$+$

We conclude then

1. f is decreasing on $(-\infty, 3)$
2. f is increasing on $(3, \infty)$.
3. f has a local minimum value at $x = 3$
4. f has no local maximum $\checkmark$


$F(x) = x^4 - 4x^3$
25
20
15
10
5
0
−5
−10
−15
−20
−25
−30
−5
−4
−3
−2
−1
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16

Study of Concavity: use f''

• $f'(x) = 4x^3 - 12x^2$ is differentiable on $\mathbb{R}$ because it is a polymonial, hence, 3) and 4) will be analized through f'':

$$\boxed{f''(x)} = 12x^2 - 24x \ \boxed{= 12x(x-2)}$$

Then, the candidates to be inflection points are $x = 0$ and $x = 2$

Concavity

	$(-\infty, 0)$	0	$(0, 2)$	2	$(2, \infty)$
x	$-$	0	$+$	0	$+$
$(x-2)$	$-$	-2	$+$	0	$+$
$f''(x)$	$+$	0	$-$	0	$+$

We conclude then

1. f has positive concavity on $(-\infty, 0)$ and on $(2, \infty)$
2. f negative concavity on $(0, 2)$.
3. f has inflection points at $x = 0$ and at $x = 2$ ✓

Asymptotes

We analize the existence of horizontal asymptotes: for this, we need to compute both limits at $\pm\infty$.

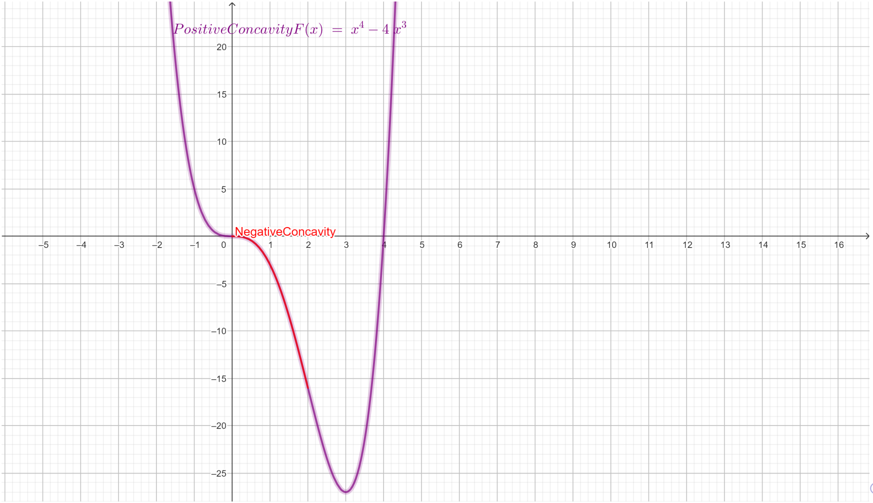
$$\lim_{x\to\pm\infty} f(x) = \lim_{x\to\pm\infty} (x^4 - 4x^3) = \lim_{x\to\pm\infty} \underbrace{x^4}_{\to+\infty} \underbrace{\left(1 - \frac{4}{x}\right)}_{\to 1} = +\infty$$

$\Rightarrow$ There is no horizontal asymptote.

- There is no vertical asymptote, since $Dom(f) = \mathbb{R}$ ✓.

All the collected information

allows to sketch the graph of the curve



L'Hospital's Rule

Suppose f and g are differentiable on an open interval (a, b) that contains c (except possibly at c). Suppose that

- $\lim_{x\to c} f(x) = 0$ and $\lim_{x\to c} g(x) = 0$, or that

- $\lim_{x\to c} f(x) = \infty$ and $\lim_{x\to c} f(x) = \infty$.

This means that f an indeterminate form of type $0/0$ or ∞/∞. Then

$$\lim_{x\to c} \frac{f(x)}{g(x)} = \lim_{x\to c} \frac{f'(x)}{g'(x)}$$

if the limit on the right side exists.

Attention: the limit on the right is not the derivative of the quotient

Proof of L'Hospital's Rule for f', g' continuous

Proof. Assume that f and g are differentiable on (a, b) that contains c $\lim_{x\to c} f(x) = 0 = \lim_{x\to c} g(x)$ and suppose f', g' continuous. Then

$$\lim_{x\to c} \frac{f(x)^{\to 0}}{g(x)^{\to 0}} = \lim_{x\to c} \frac{f(x) - f(c)}{g(x) - g(c)} = \lim_{x\to c} \frac{\frac{f(x)-f(c)}{(x-c)}}{\frac{g(x)-g(c)}{(x-c)}} = \frac{\lim_{x\to c} \frac{f(x)-f(c)}{(x-c)}}{\lim_{x\to c} \frac{g(x)-g(c)}{(x-c)}}$$

$$= \frac{f'(c)}{g'(c)} \quad \boxed{= \lim_{x\to c} \frac{f'(x)}{g'(x)} \quad \checkmark}$$

L'Hospital's Rule for Limits at $\pm\infty$

Suppose f and g are differentiable on an open interval (a, ∞). Suppose that

- $\lim_{x\to+\infty} f(x) = 0$ and $\lim_{x\to+\infty} g(x) = 0$, or that

- $\lim_{x\to+\infty} f(x) = \infty$ and $\lim_{x\to+\infty} f(x) = \infty$.

This means that f an indeterminate form of type $0/0$ or ∞/∞. Then

$$\lim_{x\to+\infty} \frac{f(x)}{g(x)} = \lim_{x\to+\infty} \frac{f'(x)}{g'(x)}$$

if the limit on the right side exists.

The theorem is also true if f and g are differentiable on an open interval $(-\infty, b)$ and we consider the limit at $-\infty$.

Check the following limits

a) $\lim_{x \to 1} \dfrac{(x-1)^{4/3}}{\ln(x)} = 0$

b) $\lim_{x \to +\infty} \dfrac{e^{x+2}}{(x+2)^{9/5}} = +\infty$

Ex. 7. Consider $f : (-\frac{1}{3}, +\infty) \to \mathbb{R}$.
a) Find $f'(x) : x \in (-\frac{1}{3}, 0) \cup (0, +\infty)$.

b) Is f differentiable at $x = 0$? If $f'(0)$ exists, find all the $\mathbf{c} \in \mathbb{R}$ such that the tangent line to the curve $y = f(x)$ at $(0, f(0))$ is parallel to the line L: $y = 10x - 1$:

$$f(x) = \begin{cases} \dfrac{e^{\mathbf{c}x} - \mathbf{c}\ln(x+1) - 1}{(\sqrt{3x+1} - 1)} & \textit{if} \quad x \neq 0 \\ 0 & \textit{if} \quad x = 0 \end{cases}$$

a) Derivative for $x \in (-\frac{1}{3}, 0) \cup (0, +\infty)$

- For $x \in (-\frac{1}{3}, 0) \cup (0, +\infty)$ the function is differentiable, because it is a composition of differentiable functions, then, we can derivate using the rules given in the table:

$$f'(x) = \frac{\mathbf{c}e^{\mathbf{c}x} - \frac{\mathbf{c}}{(x+1)}}{(\sqrt{3x+1}-1)} - 3\frac{e^{\mathbf{c}x} - \mathbf{c}\ln(x+1) - 1}{2\sqrt{3x+1}(\sqrt{3x+1}-1)^2}$$

b) Derivative for $x = 0$

- f is continuous at $x = 0$ because $\lim_{x \to 0} f(x) = 0 = f(0)$.
- $f'(0)$ must be computed by definition because the differentiability at $a = 0$ is not proved. Use L'Hopital Rule twice:

$$f'(0) = \lim_{x \to 0} \frac{f(x) - f(0)}{x} = \lim_{x \to 0} \frac{e^{\mathbf{c}x} - \mathbf{c}\ln(x+1) - 1}{(\sqrt{3x+1} - 1)x}$$

$$\stackrel{L'H}{=} \lim_{x \to 0} \frac{\mathbf{c} \cdot e^{\mathbf{c}x} - \frac{\mathbf{c}}{(x+1)}}{\frac{3x}{2\sqrt{3x+1}} + (\sqrt{3x+1} - 1)}$$

$$\stackrel{L'H}{=} \lim_{x \to 0} \frac{\mathbf{c}^2 \cdot e^{\mathbf{c}x} + \frac{\mathbf{c}}{(x+1)^2}}{\frac{3(3x+2)}{4\sqrt{3x+1}^3} + \frac{3}{2\sqrt{3x+1}}} = \frac{\mathbf{c}(1 + \mathbf{c})}{3}$$

From the condition $f'(0) = 10$, solve $\mathbf{c} = 5$ or $\mathbf{c} = -6$

- $f'(0) = \frac{\mathbf{c}(1+\mathbf{c})}{3} = 10 \iff \mathbf{c} = 5$ or $\mathbf{c} = -6$ ✓